Shengzhou Qiang

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CS PhD student in distributed transactions & databases, seeking Summer 2027 software-engineering / research internships in database internals and infrastructure.

EDUCATION

University at Buffalo, SUNY — Ph.D. in Computer Science (advisor: Prof. Haonan Lu) *Aug 2023 – present (exp. 2028)*

Research: distributed transactions, consistency protocols, and evaluation of distributed datastores.

Nanjing University of Information Science & Technology — B.S., Information and Computing Science *2019 – 2023*

GPA 3.8/5.0 (top 5%); industry-embedded program with Neusoft Group.

RESEARCH PROJECTS

SONIC — rethinking replication-critical read-only transactions (**lead author** — latest SOSP submission advanced through two review rounds to PC-meeting discussion) *2025 – present*

- Leading a first-author project showing "replication-critical" read-only transaction designs are unnecessary: formalized the SONIC and Remote-Read theorems and built two systems demonstrating them.
- Built **Starry** by deep-modifying UW's TAPIR codebase (**300+ commits, C++**) and an extended COPS-SNOW with chain replication; validated consistency with self-built **Jepsen falsifier suites** (Clojure); contributed upstream to **Jepsen** (Debian support, CockroachDB test hardening).
- Ran experiments on CloudLab via a self-built Docker-Compose cluster testbed (SSH-CA, scalable controller + N-node topology).

Juicer — transparent tail-latency layer for strictly serializable concurrency control (in submission) *2025 – present*

- Co-building a proactive message-ordering layer in gRPC-Go interceptors that reorders transaction messages by timestamp before they reach data shards — **cutting P99 latency by up to 75% and aborts by up to 85% with zero database code changes**.
- Implemented concurrency-control protocols (2PL wound-wait/wait-die, MVTO) in a **14K-LoC Go transactional KV testbed**; built its workload generator (Zipfian, dynamic skew, read-modify-write).
- Evaluated on CloudLab (9 × 16-core / 25 Gbps) against NCC (SOSP'23 state of the art): **54% tail-latency reduction**.

EaaS — Evaluation-as-a-Service for distributed storage systems *2023 – 2025*

- Co-built the first end-to-end evaluation framework for distributed storage (C++, gRPC, Raft-replicated control plane); **top contributor with ≈47% of 1,800+ commits** in a 6-person team.
- Designed the unified workload engine (YCSB, transactional YCSB-T, interactive TPC-C) and C++/Go/Java adapters integrating **CockroachDB, Cassandra, and EPaxos-family protocols** at ≈20 LoC per system.
- Re-evaluated replication protocols across AWS regions, **refuting EPaxos's published 3× throughput claim (measured 1–1.5×)**; uncovered an EPaxos livelock at 9 replicas.

PRC — developer-intended consistency testing for distributed datastores (in submission) *2025 – present*

- Directed implementation testing that turns community-known correctness pitfalls (e.g., timestamp inversion) into targeted tests, complementing random-exploration tools like Jepsen; evaluated the P formal-modeling framework.

EXPERIENCE

Neusoft Group — Software Technology Trainee (industry-embedded B.S. program) *2020 – 2023*

Built Hadoop-ecosystem pipelines: Hive warehouse, Storm streaming analytics, Spark batch jobs (Java).

PUBLICATIONS

S.-Z. Qiang et al., "Instantaneous cross-correlation function type of WD based LFM signals analysis via output SNR inequality modeling," *EURASIP J. Adv. Signal Process.*, 2021 (**first author**); 6 further co-authored papers (*Optik* ×3, *Physics Letters A*, *Math. Problems in Eng.*, *Chinese J. Electronics*), 2021–2022.

SKILLS

Languages	C++, Go, Python, Java, Clojure, Shell, LaTeX
Systems	distributed transactions & consistency (2PL/OCC/MVTO/NCC, Paxos/Raft/EPaxos, TAPIR), CockroachDB internals, gRPC, YCSB/TPC-C benchmarking, Jepsen, P formal modeling
Infra	Docker-Compose cluster orchestration (3 self-built SSH-CA testbeds), CloudLab, AWS, Linux, GitHub Actions
Other	Kaggle Foursquare Location Matching — top 10% (103/1079)